HW7

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3.4 - 15

$$B_1 = \left\{ \begin{bmatrix}1\\2\\1\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix} \right\}, \qquad B_2 = \left\{ \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}-1\\1\\-1\end{bmatrix}, \begin{bmatrix}2\\1\\-1\end{bmatrix} \right\}, \qquad [\mathbf{v}]_{B_1} = \begin{bmatrix}-1\\0\\2\end{bmatrix}.$$

we can do the row reduce method to find P:

$$\left[\begin{array}{ccc|ccc} 0 & -1 & 2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 1 & 1 \\ 0 & -1 & -1 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 2 & 1 & 1 \\ 0 & -1 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_3 - R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 2 & 1 & 1 \\ 0 & -1 & 2 & 1 & 0 & 0 \\ 0 & 0 & -3 & 0 & 1 & 0 \end{array}\right]$$

$$\xrightarrow[-R_2]{-\frac{1}{3}R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & -2 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{3} & 0 \end{array}\right] \xrightarrow{R_2 + 2R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & 0 & -1 & -\frac{2}{3} & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{3} & 0 \end{array}\right]$$

$$\xrightarrow{R_1 - R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 2 & 1 \\ 0 & 1 & 0 & -1 & -\frac{2}{3} & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{3} & 0 \end{array}\right]$$

so

$$[I]_{B_1}^{B_2} = \begin{bmatrix} 3 & 2 & 1 \\ -1 & -\frac{2}{3} & 0 \\ 0 & -\frac{1}{3} & 0 \end{bmatrix}.$$

check column 1: $3\begin{bmatrix}0\\1\\0\end{bmatrix} - \begin{bmatrix}-1\\1\\-1\end{bmatrix} + 0\begin{bmatrix}2\\1\\-1\end{bmatrix} = \begin{bmatrix}1\\2\\1\end{bmatrix}$, which is the first vector of B_1. (column 3 is free because the third vector of B_1 is the first vector of B_2.)

then

$$[\mathbf{v}]_{B_2} = [I]_{B_1}^{B_2}[\mathbf{v}]_{B_1} = \begin{bmatrix} 3 & 2 & 1 \\ -1 & -\frac{2}{3} & 0 \\ 0 & -\frac{1}{3} & 0 \end{bmatrix} \begin{bmatrix}-1\\0\\2\end{bmatrix} = \begin{bmatrix}-3+0+2\\1+0+0\\0+0+0\end{bmatrix} = \begin{bmatrix}-1\\1\\0\end{bmatrix}.$$

check: both coordinate vectors build the same $\mathbf{v}$:

$$-1\begin{bmatrix}1\\2\\1\end{bmatrix} + 2\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}-1\\0\\-1\end{bmatrix}, \qquad -1\begin{bmatrix}0\\1\\0\end{bmatrix} + 1\begin{bmatrix}-1\\1\\-1\end{bmatrix} = \begin{bmatrix}-1\\0\\-1\end{bmatrix}.$$